

MODEL QUESTION PAPER

TED(10)5009

(Revision2010)

Sixth Semester Diploma Examination in Engineering

Design of prestressed Concrete &masonry walls

Maximum Marks : 100

Time : 3 hours

(Note: IS 456-2000 and IS 1905-1987 are permitted)

Part A

I Answer the following questions in one or two sentences. Each question carries 2 marks.

1. What is critical neutral axis?
2. Define balanced section?
3. What is a tendon?
4. What is stiffening coefficient?
5. . what is effective length OF masonry walls ? (5X2=10)

Part B

II Answer any five of the following .Each question carries 6 marks

1. Derive an expression for the depth of critical neutral axis of singly reinforced rectangular section.
- 2.Explain working stress method of design.
3. List the advantages of pre stressed concrete?
4. Why are high tension steel wires used for prestressing?
5. Explain the loss of prestress due to creep of concrete.
- 6.Explains the permissible shear stress for masonry walls?
- 7.Write notes on Panel wall and partition wall. (5X6=30)

Part C

(Answer one full question from each unit. Each question carries 15 marks .Missing data may suitably assumed.)

UNIT I

III a Derive an expression for moment of resistance of a singly reinforced RCC section using M25 concrete and Fe 415 steel.

b. A rectangular beam 250mm wide has an effective depth of 460mm and is reinforced with 3 bars of 16mm diameter in the tensile zone. Find the moment of resistance of the section with M20 concrete and Fe 415 steel. 8

OR

IV a. Design a RC rectangular beam to carry a superimposed load of 12kN/m over a span of 6m. Use M20 concrete and Fe 415 steel. 8

b. A rectangular beam 250mm X 500mm deep has 4 bars of 20mm diameter Fe415 grade steel at an effective cover of 40mm. The beam is subjected to a bending moment of 60kNm. Find the actual stresses in concrete & steel. Concrete is M20 grade. 7

Unit II

V a. Explain post tensioning. 7

b. Describe the different classification of prestressed concrete. 8

OR

VI a. How are prestressed members made by the Hoyer system? 8

b. Explain the basic principles involved in the analysis of prestressed concrete structures. 7

UNIT III

VII a. Describe the loss of pre stress due to elastic shortening of concrete. 7

b. A prestressed concrete beam 400mm X 700mm in section has a span of 7m and is subjected to a uniformly distributed load of 15kN/m including the selfweight of the beam. The prestressing tendons are located along the longitudinal centroidal axis and provide an effective prestressing force of 840 kN. Determine the extreme fibre stresses in concrete at the mid span section. 8

OR

VIII a. A prestressed concrete beam of uniform rectangular cross section and span 18m supports a distributed load of 20kN/m excluding the weight of the beam. Determine the suitable dimensions of the beam and calculate the area of the tendons and their position. The permissible stresses are 14N/mm² for concrete & 1025N/mm² for the tendons. 15

UNIT IV

IXa. Write notes on load bearing walls and non load bearing walls. 7

b. Explain the analytical method of design of masonry walls.

8

OR

X a. A masonry wall carrying an axial load of 10.5 KN/m is of 3.6m effective length. It is not braced by cross walls. The effective height of the wall is 2.5m. Design the masonry wall. 15
